

Practice problems #1

1. Consider the following statement:

For an integer n , if n^2 is even, then n is even.

- (a) Explain why the indirect proof would work better to prove this statement.
- (b) Prove the statement by contradiction.
- (c) Prove the statement by contrapositive.

Answers or Hints are on the second page.

Hints:

#1(a): Is n easy, or n^2 easy?

#1(b): Suppose not. Then, n is ..., so n^2 is ... But n^2 is This is a contradiction.

#1(c): Will show: If n is ..., then n^2 is ...